

Response to first drug trial predicts outcome in childhood temporal lobe epilepsy

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Article abstract—*Objective:* To construct a clinical prediction model for the early identification of children destined to develop refractory temporal lobe epilepsy (TLE) 2 years after epilepsy onset. *Methods:* Patients with TLE between 1 and 18 years old seen in the Division of Neurology at Children's Hospital of Philadelphia during 1999 were identified through billing records and chart review. Data were abstracted independently on 5 candidate predictor variables for refractory TLE and on seizure frequency outcome at 2 years after epilepsy onset. *Results:* One hundred twenty patients met inclusion criteria and had at least 2 years of follow-up. Forty-five of 120 patients (37.5%) had refractory TLE at 2 years after onset, and 75 of 120 (62.5%) were seizure free. Three significant predictors of refractory TLE were found on bivariate analysis: an early risk factor for epilepsy (risk ratio = 3.5 [95% CI 2.2, 5.6]), temporal lobe abnormality on MRI scan (2.9 [95% CI 1.9, 4.6]), and failure of the first antiepileptic drug (AED) trial (16.5 [95% CI 6.3, 43.9]). Logistic regression indicated that the best model to predict refractory TLE contained only the variable "failure of first AED trial," with a positive predictive value of 0.89 (95% CI 0.76, 0.96) and negative predictive value of 0.95 (95% CI 0.87, 0.99) to predict "refractory TLE" at 2 years. *Conclusions:* Failure of first AED trial accurately predicts refractory TLE at 2 years after onset, based on retrospective cohort data in children. If verified prospectively and with longer follow-up, this finding should support earlier consideration of surgical options.

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Intractable epilepsy can lead to cognitive decline, poor socialization, depression, and increased mortality.^{1,2} The psychosocial consequences of refractory epilepsy may be especially disabling during adolescence.³ Usual treatment options for refractory temporal lobe epilepsy (TLE) include additional antiepileptic drug (AED) trials or consideration for epilepsy surgery (temporal lobectomy). However, not all children with TLE develop refractory seizures; some children become seizure free with AED and may eventually enter remission.

The early identification of children destined to develop refractory TLE might justify earlier surgical therapy. TLE is distinct among the different types of epilepsy because surgery offers a 70% to 80% chance of a seizure free outcome.^{4,5} However, surgery is invasive, irreversible, and is associated with memory dysfunction in some patients.⁶

We sought to develop a clinical prediction model for the early identification of children destined to have refractory TLE 2 years after epilepsy onset. A model to accurately identify children at high risk for refractory TLE, based on initial clinical findings, would allow for earlier counseling about treatment options, including the possibility of earlier surgery in appropriate patients. A clinical prediction model in-

cludes variables obtained from the history, physical examination, or diagnostic tests and provides the probability of an outcome or suggests a diagnostic or therapeutic course of action.⁷⁻⁹ We hypothesized that a prediction model with an accuracy of at least 90% to predict refractory TLE at 2 years after onset could be constructed based on information available early in the course of childhood TLE.

Methods and patients. *Subjects.* This retrospective cohort study sought to identify every patient between 1 and 18 years old with TLE who was seen for an outpatient visit in the Division of Neurology at Children's Hospital of Philadelphia (CHOP) during 1999 with onset of epilepsy after 1996. Clinical settings included an urban university-based outpatient practice and six suburban satellite outpatient practices, which encircle the city of Philadelphia. Potential patients were identified through centralized billing records, using *International Classification of Diseases, Ninth Edition* (ICD-9) codes 345.4x (partial epilepsy with impairment of consciousness), 345.8x (other forms of epilepsy), 345.9x (epilepsy unspecified), and 780.3 (convulsions). All outpatient charts with ICD-9 codes 345.4x and a visit during 1999 were reviewed for patients meeting inclusion criteria. A sample of 250 charts with the other codes was reviewed to ensure that eligible patients were not missed because of nonspecific coding.

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Eligible subjects for this study had a history of at least two complex partial seizures with clinical seizure semiology suggesting temporal lobe onset, such as staring, unresponsiveness, automatisms, or auras. Clinical seizure semiology and predictor variable data were abstracted from the patient's initial outpatient visit. Temporal lobe EEG or MRI abnormalities were noted but were not specifically required for inclusion. Patients with benign rolandic epilepsy, any other epilepsy syndrome distinct from TLE, documented seizure onset outside of the temporal lobe, or a history of refractory TLE before initial referral to CHOP were excluded.

The study sample consisted of patients referred to and followed by a pediatric neurologist and was not a population-based sample. Patients with childhood TLE who were managed by their primary care physician, an adult neurologist, or a non-CHOP pediatric neurologist were not captured in this study. The CHOP suburban satellite specialty centers examine large numbers of outpatients, without requiring a visit to the tertiary care hospital in Philadelphia. Patients examined at these centers were included in the study, allowing for an appropriate mix of disease severity.

Data collection. Based on a literature review,¹⁰⁻¹⁷ five candidate predictor variables for refractory TLE were determined before data extraction began. The predictor variables were chosen because they were well specified, clinically appropriate, reliably measured between observers, likely to be reproducible in multiple centers, and dependably available through retrospective chart review. The five predictor variables were age at epilepsy onset,^{10,11} early risk factors for epilepsy,¹¹⁻¹⁴ family history of epilepsy in a first-degree relative,¹⁵ temporal lobe abnormality on MRI scan,¹⁶ and outcome of the first AED trial.¹⁷ Age at epilepsy onset was considered the age at first unprovoked seizure. Early risk factors for epilepsy included a history before 5 years of age of any of the following: febrile seizures,¹⁴ CNS infection such as meningitis or encephalitis,^{13,14} head trauma with loss of consciousness for greater than 30 minutes,¹²⁻¹⁴ or neonatal seizures.¹¹ Data on individual risk factors were abstracted, and a combined variable was created. Temporal lobe MRI abnormalities included findings such as abnormal signal intensity, atrophy, or foreign tissue lesions in the temporal lobe(s). MRI abnormalities outside the temporal lobe were noted, but an abnormality in the temporal lobe(s) must have been noted in the official MRI report to be considered abnormal. An AED trial was considered a failure if seizures persisted at the maximally tolerated dose. A standard algorithm was used to assess the results of AED trials (figure). Idiosyncratic reactions and medication noncompliance, as determined by the treating neurologist, were not considered AED trial failures. The outcome of interest was "refractory TLE," defined as the persistence of any seizures involving impairment of consciousness between 18 and 24 months after epilepsy onset and despite at least 2 maximally tolerated AED trials. Patients with persistent or breakthrough seizures on less than maximally tolerated AED therapy were classified as having "suboptimal AED therapy" and were not included in the data analysis. Outcome data as of December 31, 2000, were used in the analysis. Predictor variables and outcome were abstracted independently.

Data were entered into a password-protected Microsoft

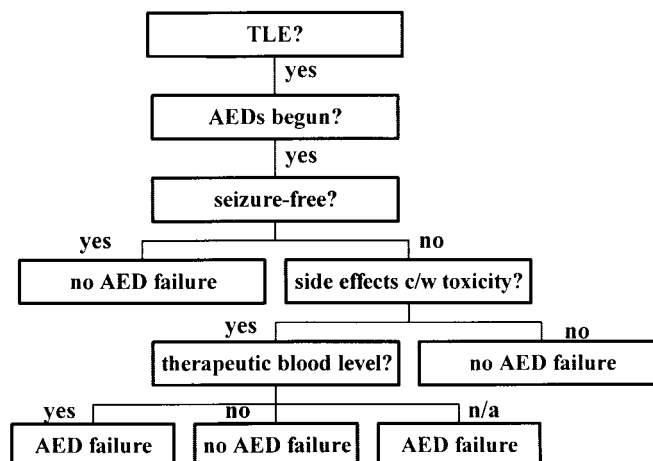


Figure. Schematic of the standard algorithm used to assess the results of antiepileptic drug (AED) trials. TLE = temporal lobe epilepsy.

Access 97 database (Microsoft, Redmond, WA). Patient-identifying information was replaced with study-specific identifiers. Statistical analyses were performed using STATA 6.0 (STATA Corp., College Station, TX). Initially, bivariate analyses were conducted to evaluate each of the five candidate predictors versus the outcome of "refractory TLE." The variable "age at epilepsy onset" was considered as both a continuous variable and a categorical variable with four groups. The association of each predictor versus outcome was calculated from 2×2 tables, using a χ^2 test of association. A multivariable analysis followed using logistic regression techniques. Positive predictive value (PPV) was defined as the percentage of patients with positive predictor variables who had refractory TLE at 2 years after epilepsy onset. Negative predictive value (NPV) was defined as the percentage of patients with negative predictor variables who were seizure free. Exact binomial 95% CI were computed.

Results. Overall, 1,846 outpatient records with ICD-9 codes 345.4x were reviewed. Of these, 162 patients (8.7%) met inclusion criteria, and 120 patients had at least 2 years of follow-up data from the onset of epilepsy. Of the remaining 42 patients, 26 had epilepsy for less than 2 years (outcome of interest too early to classify), 6 were lost to follow-up, 4 underwent temporal lobectomy before 2 years, and 6 had "suboptimal AED therapy." Of the six patients with "suboptimal AED therapy," five had breakthrough seizures while receiving less than maximally tolerated AED dosages and one was considered noncompliant by the treating neurologist. No eligible patients were identified upon review of the 250 charts with less specific ICD-9 codes.

Forty-five of 120 patients (37.5%) had refractory TLE at 2 years after epilepsy onset, and 75 of 120 (62.5%) were seizure free. The overall composition of the cohort and outcome did not differ based on sex: 63 of 120 patients (52.5%) were boys, including 22 of 45 (49%) of patients with refractory TLE.

Bivariate analyses of the five predictor variables versus outcome are found in table 1. Significant predictors of refractory TLE based on bivariate analysis included the

Table 1 Bivariate comparisons of predictors of refractory TLE

Predictor variable	No. of subjects	No. (%) with refractory TLE at 2 y	Unadjusted risk ratio (95% CI)
Early risk factor for epilepsy			
Yes	38	28 (73.7)	3.5 (2.2, 5.6)
No	82	17 (20.7)	
Febrile seizures			
Yes	27	22 (81.4)	3.3 (2.2, 4.9)
No	93	23 (24.7)	
Temporal lobe MRI abnormality			
Yes	37	26 (70.3)	2.9 (1.9, 4.6)
No	80	19 (23.7)	
Failure of first AED trial			
Yes	46	41 (89)	16.5 (6.3, 42.9)
No	74	4 (5.4)	
Family history of epilepsy			
Yes	5	2 (40)	1.1 (0.3, 3.2)
No	115	43 (37.3)	

Mean $\pm$ SD age at epilepsy onset was 6.4 ± 4.1 years in 45 patients with refractory TLE and 7.4 ± 3.9 y in 75 patients who were seizure free. This variable was analyzed as both a categorical variable using a χ^2 test and a continuous variable using a Student's *t*-test and was not significantly associated with refractory TLE.

TLE = temporal lobe epilepsy; AED = antiepileptic drug.

presence of an early risk factor for epilepsy, temporal lobe abnormality on MRI scan, and failure of the first AED trial. Among the 38 patients with an early risk factor for epilepsy, 28 had febrile seizures, 7 had CNS infections, 2 had neonatal seizures, and 1 had significant head trauma. Only the subgroup with febrile seizures was large enough for separate analysis, and a history of febrile seizures was a significant predictor of refractory TLE. MRI scans were obtained in 117 of 120 patients (98%). Among the 37 patients with a temporal lobe MRI abnormality, 18 had abnormal signal intensity without other findings, 15 had foreign tissue lesions (7 cysts, 3 gangliogliomas, 3 unspecified mass lesions, 2 hemangiomas), 8 had changes consistent with mesial temporal sclerosis, and 6 had atrophy. Of the 120 patients, the medication chosen for the first AED trial was carbamazepine for 102 (85%), phenobarbital for 8 (7%), phenytoin for 5 (4%), valproic acid for 4 (3%), and gabapentin for 1 (1%).

Age at epilepsy onset was evaluated as both a continuous variable and a categorical variable using 4 categories: less than 2 years, 2 to 6.9 years, 7 to 12 years, and greater than 12 years. Regardless of the analytical method, age at onset was not associated with refractory TLE. Family history of epilepsy in a first-degree relative also was not associated with refractory TLE, although the number of patients with a positive family history was quite small.

Logistic regression techniques were then used in a mul-

Table 2 Results of multivariate logistic regression

Predictor variable	Regression coefficient* (95% CI)	OR for refractory TLE (95% CI)
Early risk factor for epilepsy: yes	1.9 (0.4, 3.4)	6.6 (1.4, 30.1)
Temporal lobe MRI abnormality: yes	1.7 (0.2, 3.2)	4.1 (1.2, 23.9)
Failure of first AED trial: yes	4.5 (3.0, 6.1)	93.8 (20.1, 437.8)

* The regression coefficients provide information about the relative weighing of the predictor variables in predicting outcome. As the coefficients for "early risk factor for epilepsy" and "temporal lobe MRI abnormality" are relatively small and the 95% CI span 1.0, these variables do not add to the accuracy of the prediction model. The OR for each predictor variable is derived from the regression coefficient, but the regression coefficient is used in the construction of the prediction model.

TLE = temporal lobe epilepsy; AED = antiepileptic drug.

tivariate analysis (table 2) to identify independent predictors of refractory TLE. The dependent variable was the outcome "refractory TLE" and the three independent variables were an early risk factor for epilepsy, temporal lobe MRI abnormality, and failure of the first AED trial. Further analysis of the three individual predictor variables (table 3) indicated that all of the predictive power of the model was contained within the variable "failure of first AED trial." Including only the variable "failure of first AED trial" resulted in a PPV of 0.89 (95% CI 0.76, 0.96) and NPV of 0.95 (95% CI 0.87, 0.99) to predict refractory TLE, defined as persistent seizures at 2 years after epilepsy onset and despite at least 2 maximally tolerated AED trials. Thus, the best prediction model to identify refractory TLE included only one variable: failure of first AED trial.

Two secondary analyses were performed. First, we examined whether patients initially diagnosed with TLE based on seizure semiology plus an EEG or MRI abnormality differed from patients diagnosed using seizure semiology alone. Eighty-eight of the 120 patients (73%) had a focal EEG or MRI abnormality in the temporal lobe, and the remaining 32 patients (27%) were diagnosed using seizure semiology alone. Outcome did not differ between the 2 groups; 35 of 45 patients (78%) with refractory TLE were initially diagnosed based on semiology plus EEG or MRI compared with 53 of 75 patients (71%) who were seizure free (risk ratio = 1.3 [95% CI 0.72, 2.26]). We also examined whether the outcome of patients initially treated with carbamazepine differed from that of patients treated with other AED. Outcome did not differ between the 2 groups; 38 of 45 patients (84%) with refractory TLE were initially treated with carbamazepine compared with 64 of 75 patients (85%) who were seizure free (risk ratio = 1.04 [95% CI 0.56, 1.96]).

Discussion. We found that a clinical prediction model containing the single variable "failure of first AED trial" can predict the outcome "refractory TLE" at 2 years after epilepsy onset with a PPV of 0.89 (95% CI 0.76, 0.96) and an NPV of 0.95 (95% CI 0.87,

Table 3 PPV, NPV, and accuracy of models to predict refractory TLE

Variables in model (no. of variables)	PPV (95% CI)	NPV (95% CI)	Accuracy (95% CI)
Early risk factor for epilepsy (1)	0.74 (0.57, 0.87)	0.79 (0.69, 0.87)	0.78 (0.69, 0.85)
Temporal lobe MRI abnormality (1)	0.70 (0.53, 0.84)	0.76 (0.65, 0.85)	0.74 (0.65, 0.82)
Early risk factor and MRI abnormality (2)	0.70 (0.57, 0.82)	0.92 (0.82, 0.97)	0.81 (0.73, 0.88)
Early risk factor, MRI abnormality, and failure of first AED trial (3)	0.89 (0.76, 0.96)	0.94 (0.86, 0.98)	0.92 (0.86, 0.96)
Failure of first AED trial (1)	0.89 (0.76, 0.96)	0.95 (0.87, 0.99)	0.93 (0.86, 0.97)

PPV = positive predictive value; NPV = negative predictive value; TLE = temporal lobe epilepsy; AED = antiepileptic drug.

0.99), based on retrospective cohort data in children. Thus, 89% of children with TLE who had persistent seizures after a single maximally tolerated AED trial subsequently developed refractory TLE 2 years after epilepsy onset. Alternatively, 95% of children with TLE who were seizure free after the first AED trial were seizure free 2 years after epilepsy onset.

The prediction models summarized in table 3 suggest that the ability to predict outcome accurately at 2 years after TLE onset steadily increases through the initial months after diagnosis and initiation of treatment. For example, at the initial clinical visit, obtaining a history of early risk factors for epilepsy (defined in this study as febrile seizures, CNS infection, head trauma with loss of consciousness for >30 minutes, or neonatal seizures) allowed for an accurate prediction of 2-year outcome in 78% of cases. After the MRI result was known, combining information about early risk factors and MRI improved the predictive accuracy slightly to 81%. However, after the results of the first AED trial were known, the predictive accuracy improved to 93%. Furthermore, once the results of the first AED trial were known, the other factors no longer contributed to the accuracy of the prediction model.

In this study, the classification of “refractory TLE” at 2 years after onset required at least two AED trial failures in maximally tolerated dosages. The results indicate that if the first AED trial failed, additional AED therapy resulted in complete seizure control in only 5 of 46 children (11%) with TLE. Studies of partial epilepsy in adults have reported that patients who fail 1 AED trial have only a 12% to 14% chance of complete seizure control with other AED monotherapy^{17,18} and a 3% to 11% chance with AED polytherapy.^{17,19} Consequently, many adult epilepsy centers define medical intractability as ongoing seizures despite two maximally tolerated AED trials.²⁰ The issue is more complicated in children. A population-based study in Nova Scotia, for example, found that remission of seizures occurred in 42% of children who failed their first AED trial,²¹ although this study included patients with a variety of seizure types. The results of the current study suggest that children with TLE who have a poor response to initial AED therapy share the more ominous prognosis of adults with partial epilepsy rather than the more

favorable prognosis of children with epilepsy in general.

The current study found an association between a history of febrile seizures and refractory TLE (risk ratio 3.3 [95% CI 2.2, 4.9]). The relation between febrile seizures and TLE is complex and depends on the study design. Surgical series have found an association between febrile seizures and TLE.²² In contrast, population-based studies have failed to find an association between febrile seizures and subsequent development of localization-related epilepsy or TLE.²³⁻²⁴ However, population-based studies have not examined whether, within the group of patients with TLE, febrile seizures were associated with refractory TLE, which is the association found in the current study. Such an association does not demonstrate that febrile seizures cause refractory TLE, as both febrile seizures and subsequent refractory TLE could be markers for temporal lobe structural abnormalities like cortical dysplasia.²⁵

Four previous studies have examined potential predictors of outcome for childhood epilepsy in general, but not for TLE specifically. Two studies looked at predictors of remission, and two studies examined predictors of intractability. A population study of 504 patients in Nova Scotia found that age of <12 years at onset, normal intelligence, no prior neonatal seizures, and fewer than 21 seizures before treatment were “reasonably accurate” predictors of remission.¹¹ A population study of 245 patients in Finland found that cryptogenic etiology, early response to drug therapy, low initial seizure frequency, and seizure type were independent predictors of remission.²⁶ A hospital-based study of 172 patients in the United States found that a history of infantile spasms, early age at onset, remote symptomatic etiology, and history of status epilepticus were independent predictors of intractability.¹⁰ Finally, a hospital-based study of 466 children in the Netherlands developed a regression model to predict outcome in newly diagnosed epilepsy. The predictive model was correct in 66% of children in whom a “poor” outcome was predicted and 79% of children in whom a “not poor” outcome was predicted.²⁷ None of these studies were “syndrome specific” and thus were limited by the great heterogeneity of childhood epilepsy. The development and validation of a clinical prediction model

unique to childhood TLE, given its excellent response to surgical therapy, would be of significant clinical importance.

The efficacy of temporal lobectomy is well established, but patient referral for surgery is often delayed. A recent adult series of 89 patients reported a mean age at epilepsy onset of 13.1 years but mean age at surgery of 31.9 years, resulting in a mean duration of epilepsy of 18.8 years before surgery.⁴ Although surgery clearly must be avoided in children destined to become seizure free within a reasonable period of time, excessive delay in pursuing effective surgical therapy risks permanent psychosocial disability.²⁸ This study indicates that failure of the first AED trial in a child with TLE is a sign indicating cause for concern and perhaps should prompt discussion of surgery as a potential option if, in fact, at least one additional AED trial is unsuccessful.

There are several limitations to this study. The retrospective nature of the study risks bias: both selection bias and information bias. Selection bias results in a cohort study if the choice of an exposed or nonexposed subject is related to the outcome of interest. In this study, selection bias would be a problem if exposed subjects (those with positive predictors for refractory TLE) who actually develop “refractory TLE” were more (or less) likely to be enrolled and/or to complete follow-up than exposed subjects who became seizure free. Selection bias is minimized by enrolling all possible exposed and nonexposed patients with TLE and by avoiding losses to follow-up. Because data are available on all TLE patients seen during 1999 and only 6 of 120 (5%) were lost to follow-up, the risk of significant selection bias is slight.

Information bias could result from patient misclassification or inaccurate data collection. Patient misclassification is a potential concern. It is not completely certain that all of the patients in the study truly have TLE. Even with strict inclusion criteria, it is possible that some patients have other forms of partial epilepsy that may mimic TLE, such as frontal or occipital lobe epilepsy with rapid propagation to the temporal lobe. We have inferred the area of seizure onset based on clinical semiology. The inclusion criteria included the typical clinical symptoms of temporal lobe seizure onset: staring, unresponsiveness, automatisms, and possible aura. These inclusion criteria are widely accepted by neurologists and have been used in other studies of TLE.¹⁶ Patients with clear evidence of seizure onset outside the temporal lobe were not included.

Inaccurate data are also a concern. If families of patients who have “refractory TLE” recalled predictors more accurately, then this would result in differential bias and be a serious concern. This is unlikely to be a problem because history regarding the predictors is taken at the child’s initial visit—before it is known whether the child will have intractable seizures or not. In addition, the predictor variables were clear enough that all parents were likely to

recall them accurately. Virtually all children received a MRI scan, so this is an unlikely source of bias. Finally, the accuracy of patient-reported or parent-reported outcome data on seizure frequency is a concern. This is a universal problem in clinical epilepsy research. The binary seizure classification system (seizure free or not seizure free) used in this study is likely more accurate than systems that depend on reporting number of seizures per month. Regardless, if either predictors or seizures were under reported (which is much more likely than being over-reported), this would have resulted in decreased power to detect a true difference between patient groups and would have biased the results of the prediction model toward the null.

Confounding is less of an issue in the construction of a prediction model, as we are interested in predicting outcome and less interested in the effect size of individual risk factors. The predictor variables certainly do not represent the only risk factors for “refractory TLE.” Other factors not available or too imprecise to include in a prediction model—developmental delay, mental retardation, occult pathologic abnormalities—may be important risk factors. Some data suggest that microscopic pathologic abnormalities in the temporal lobe (cortical dysplasia) may be an important risk factor for intractable TLE.²⁵ Occult dysplasia is important in the etiologic understanding of TLE but cannot be used to predict clinical outcome because it currently requires pathologic confirmation. We purposely did not choose “remote symptomatic epilepsy” as a predictor variable because it is defined differently by different authors (and by different treating neurologists) and is not believed to be sufficiently reliable or reproducible for inclusion in the model. A review of five prominent articles on the prognosis of epilepsy found that all authors defined “remote symptomatic” in a different way.^{17,27,29-31} Some authors required a structural abnormality, whereas others allowed for the presence of a static encephalopathy, which was then defined in various ways. The predictors used in the study likely serve as markers for the true risk factors, which does not take away from the clinical utility of the prediction model. The aim of the study was not to identify risk factors for refractory TLE but to predict its outcome.

The ability to generalize the results is a concern because the study sample consisted of patients referred to and followed by a pediatric neurologist and was not a population-based sample. Even if more severe cases were captured in this study, the results remain clinically useful. A clinical prediction model is not needed for the management of children with uncomplicated, easy-to-control epilepsy—the patients most likely to be missed by this study. Management questions arise when children do not respond to initial therapies, and such patients are extremely likely to be referred to a pediatric neurologist. This clinical prediction model is designed for use by pediatric neurologists, not primary care providers, so if the most benign end of the TLE spec-

trum is not sufficiently represented, the goals of the study remain intact. Overall, we believe that the study population provided a representative sample of patients with childhood TLE referred to and followed by a pediatric neurologist. The study did not attempt to make population-based estimates of the prevalence or incidence of TLE.

As carbamazepine was the first AED used in 85% of the patients, it cannot be excluded that the choice of different initial AED would have produced a different outcome. An important area for future study is whether early use of AED with different mechanisms of action improves outcome in patients with epilepsy syndromes at high risk for intractability.

This study was not designed to characterize the prognosis of childhood TLE beyond 2 years after onset. This is a challenging issue to address with a retrospective cohort study, as the number of patients lost to follow-up increase and patients undergo temporal lobectomy, thus altering the natural history of the disease. Older hospital-based studies of childhood TLE suggest a poor prognosis. A series of 100 children with TLE followed into adulthood found that only 33% were seizure free and living independently.³² Another series found only 10% of 63 children with TLE to be seizure free after 6 years of medical treatment.³³ Population-based data on the long-term prognosis of childhood TLE are simply not available.

Any clinical prediction model should be validated in a prospective fashion before being incorporated into clinical practice. If it is verified prospectively that the results of the first AED trial accurately predict 2-year outcome in children with TLE, counseling regarding earlier surgical therapy could be considered.

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